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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	PnJ
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG4 Quality Education(main); supporting SDG8 and SDG9
Project / Solution Name	FutureReady — An AI-Assisted Career Readiness Platform
Name (Leader)	Jessy Chia Jiek Xin
Email & Phone (Leader)	jimhow00@gmail.com / 011-55596939
University Affiliation(s)	Multimedia University (MMU)
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI adoption is changing how people prepare for employment, while students and early-career users may struggle to connect their current skills with realistic career pathways. They need a practical way to identify transferable skills, recognize gaps, and prioritize accessible learning rather than relying on generic career advice. The challenge aligns with SDG Target 4.4, which emphasizes relevant skills for employment and entrepreneurship.

FutureReady addresses this challenge for students, graduates, and early-career individuals exploring technology-related pathways. Its current prototype provides five team-curated career pathways, explainable skill-based comparisons, and structured learning recommendations. It also helps users interpret AI-related workforce signals cautiously: illustrative job listings are not vacancy totals, and moderate/fast AI-adoption scenarios are not forecasts.

The project aims to make the next learning step easier to identify, without presenting a match score as a hiring probability or claiming proven improvements in employment outcomes. User testing is needed to measure usability and the usefulness of recommendations.

Primary alignment: SDG 4 (Quality Education), particularly Target 4.4. Supporting potential connections: SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation and Infrastructure). These are team-defined alignments, not independently verified impact outcomes.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureReady is a deployed web-based career-readiness platform designed to help students and early-career individuals explore career pathways, identify skill gaps, and develop personalized learning plans in the AI era.

The platform consists of six pages: Home, My Skills Profile, Career Matches, My Learning Roadmap, SDG Mapping, and AI Trends & Scenarios. Users create a skills profile by entering their proficiency levels, education, experience, interests, and career goals. A rule-based matching engine compares their

skills against five team-curated career pathways informed by O*NET and ESCO, generating explainable match scores and identifying areas for improvement.

FutureReady provides structured learning roadmaps with free or low-cost resources and progress tracking. It also highlights potential contributions to SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation and Infrastructure).

The AI Trends & Scenarios feature presents referenced workforce research, illustrative job listings from Remotive, and two team-defined AI adoption scenarios. Optional AI-generated career insights are integrated through RunPod Serverless, with PostgreSQL supporting asynchronous job tracking. Public AI inference is currently disabled by default to prevent uncontrolled GPU usage, while core career guidance remains available.

The application is built using React, TypeScript, Vite, Express, PostgreSQL, and Drizzle ORM, and deployed through Replit. Replit Agent and ChatGPT assisted development, supported by official sponsor-provided Replit and RunPod credits.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technology Stack

- Frontend: React, TypeScript, Vite and Tailwind CSS for the user interface.
- Backend: Node.js and Express for API integration, career guidance and AI job management.
- Database: PostgreSQL with Drizzle ORM for asynchronous AI job tracking. User profiles and roadmap progress are stored in browser localStorage.
- Career Matching: A rule-based engine compares user skills with five team-curated career pathways informed by O*NET and ESCO references.
- External API: Remotive API provides illustrative remote job listings.
- AI Integration: RunPod Serverless supports optional AI-generated career insights using an intended Qwen3-4B-Instruct-2507 model.
- Development and Deployment: Replit for application development and hosting; GitHub for version control.

System Architecture

User → React Frontend → Express Backend → Career Matching Engine / Remotive API / RunPod Serverless

Browser localStorage stores user profiles and learning progress, while PostgreSQL stores AI job-tracking records.

Yes. FutureReady utilized official sponsor-provided Replit credits for development and deployment, and RunPod credits for GPU-based AI inference and integration testing.

For the public prototype, RunPod inference is disabled by default through a server-side feature flag to prevent uncontrolled GPU usage. Core rule-based career guidance remains fully available.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define. We focus on students, graduates, and early-career users who need to understand their transferable skills, compare possible careers, and select practical next learning steps amid uncertainty about AI adoption.

Ideate and prototype. We built six connected pages around an editable skills profile, explainable rule-based career matching, skill-gap identification, learning resources, SDG mapping, and cautious AI-adoption scenarios. Curated role/skill data and cited trend cards are distinguished from illustrative Remotive job listings; optional generative AI is not required for core guidance.

Test and deploy. The React frontend and Express API production builds and TypeScript checks passed. A public Replit prototype has been published. The next validation step is a browser-based end-to-end walkthrough, production database verification, and pilot usability testing with synthetic profiles. RunPod is disabled for public use pending durable quota protection; no verified user-impact results are claimed yet.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4, Target 4.4 (primary); SDG 8, Target 8.5 (supporting); SDG 9 (supporting potential connection). Primary: SDG 4.4 — increase youth and adults with relevant technical and vocational skills for employment, decent jobs and entrepreneurship. Supporting: SDG 8.5; potential SDG 9 connection through responsible digital innovation.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Prototype baseline: no pilot-user outcome data has yet been collected (0 completed pilot evaluations). Existing public research may motivate the problem, but a Malaysia-specific numeric workforce baseline should be added only after its source and quotation are verified.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: percentage of pilot users who complete a skills profile, view career matches, and reach an ordered learning roadmap within 10 minutes. Secondary measures: task completion rate, source-link accessibility, and users' ability to explain the match score and scenario limitations. Record timings and results during pilot testing.
Phase 2 Impact Target	

Metric Category	Your Response
Your realistic, percentage-based target by the final prototype presentation *e.g., "20% improvement in computational resource use."*	Phase 2 target (proposed, not achieved): at least 80% of pilot participants complete the profile → career match → roadmap journey within 10 minutes. Aim to check 100% of displayed learning-resource links in the pilot catalogue. These are usability targets, not evidence of improved employment outcomes.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes [/] No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[/] Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	*Type (code/dataset/model)*	*License*
React / Express / Vite / Drizzle ORM	code	MIT / Apache-2.0 (per package)
Remotive API / O*NET & ESCO references	API / reference data	Provider terms / attribution to verify
Qwen3-4B-Instruct-2507	pre-trained model	Model-specific license to verify

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes [/] No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Jessy Chia

25/10/26